



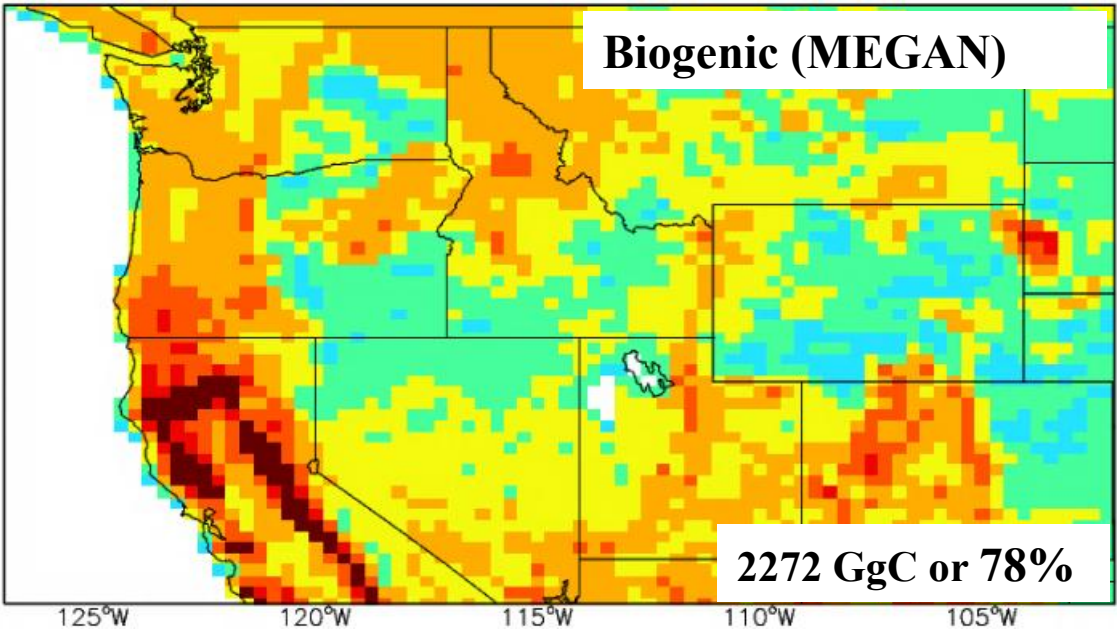
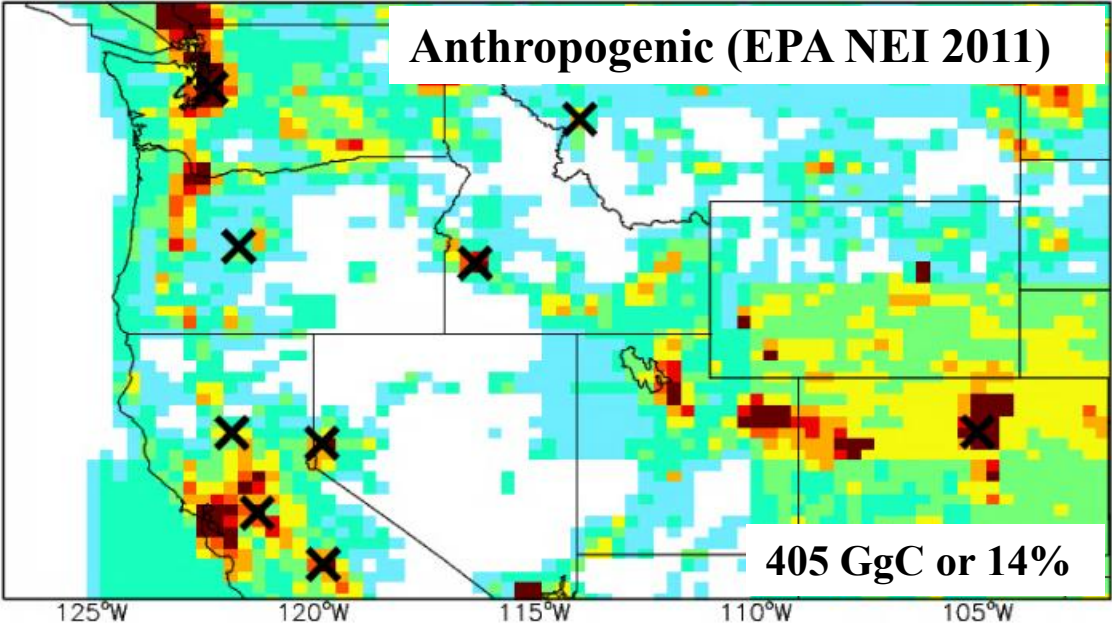
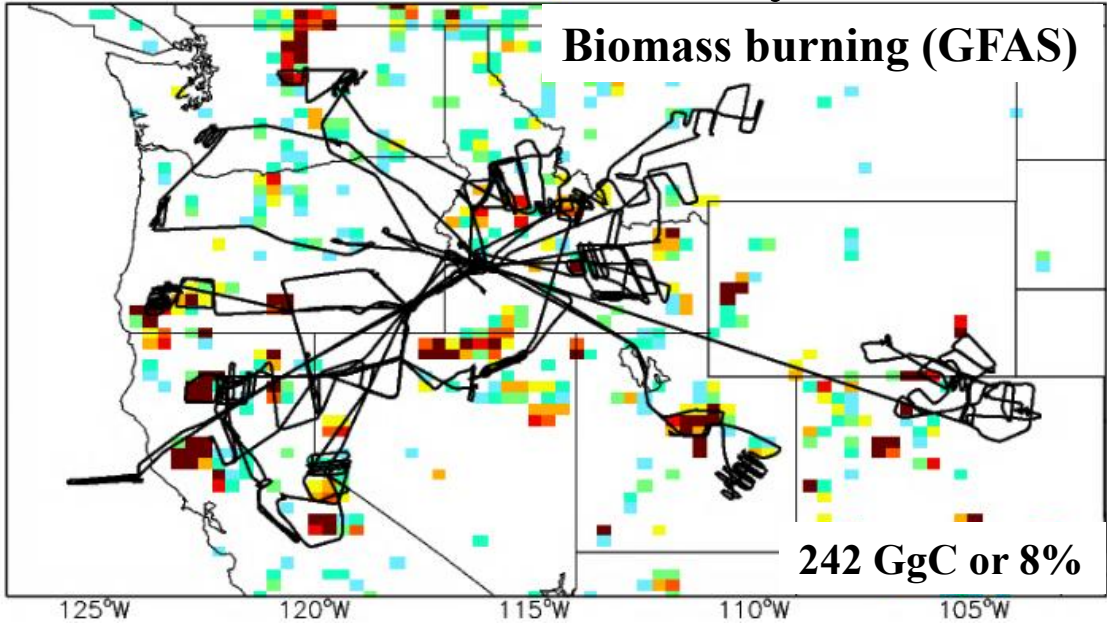
Constraining VOC emission from western US wildfires with WE-CAN and FIREX-AQ airborne observations

Lixu Jin

Group meeting

October 7th, 2022

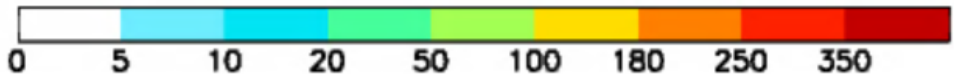
Model suggests that BB emission only account for < 10% of the VOC primary emission sources for different years and inventories.



Biomass burning/Anthropogenic



Biogenic

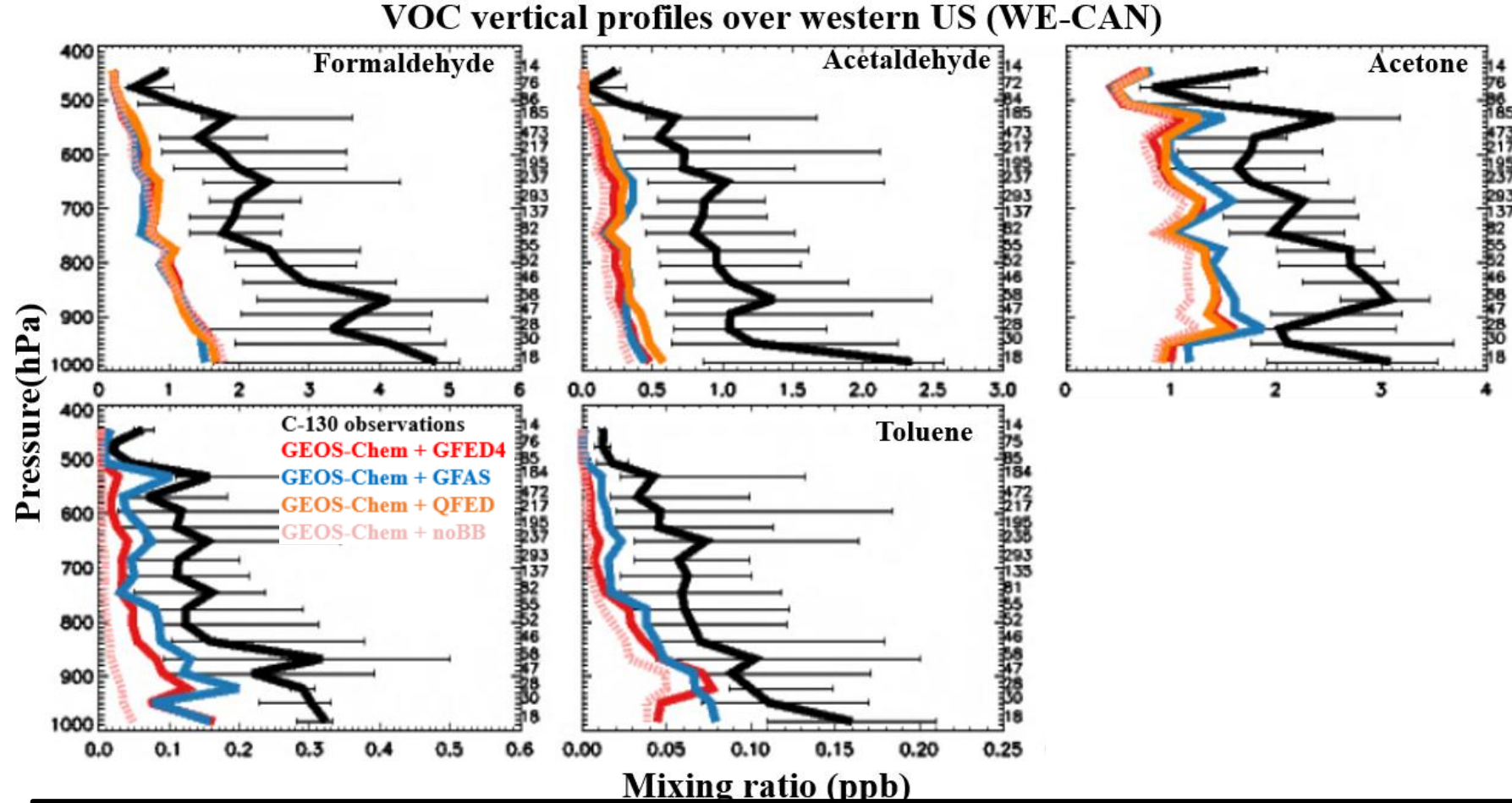


[1e12 atomsC/cm2/s]

Across 3 different BB emission inventories, the total VOCs emissions is diff by 30-40%.

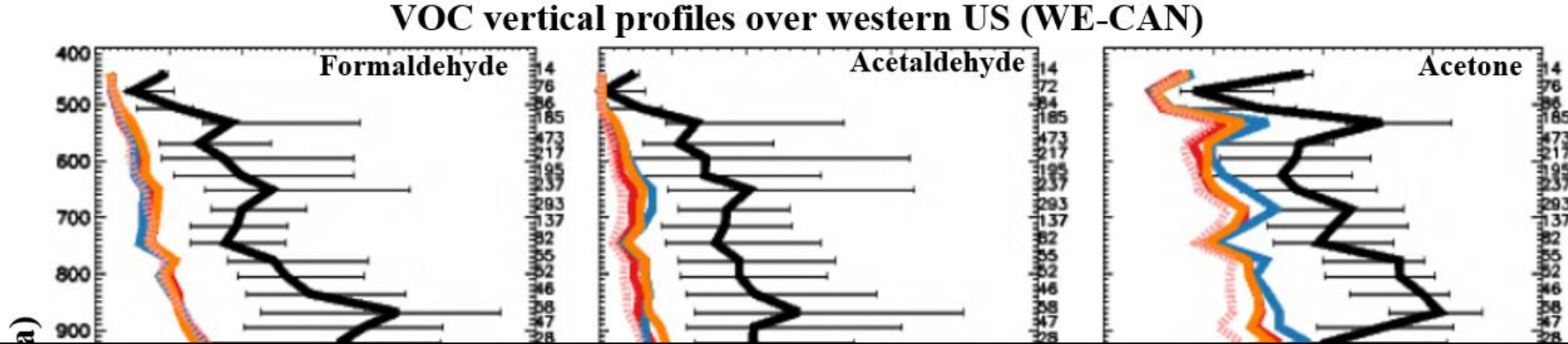
In 2019 summer, VOCs from BB emissions are ~1/6 of the 2018 (not shown), reflecting the large fire variability.

The model can predict the BB enhancements, but the emission size may be underestimated

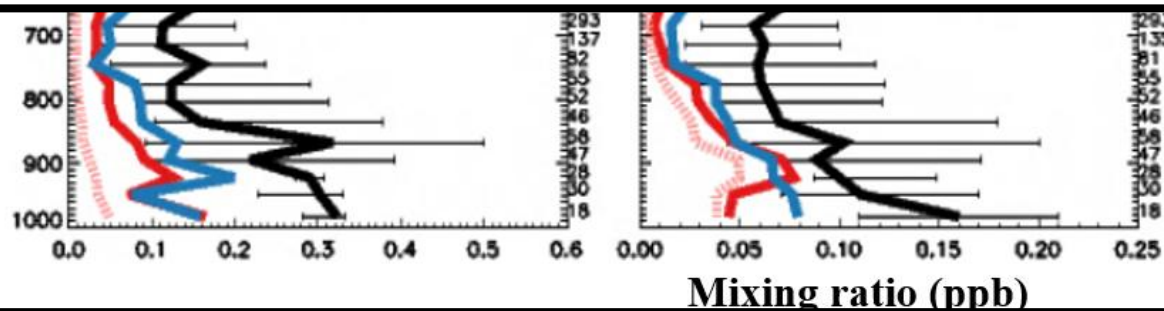


- The model with different inventories suggest similar enhancements and tend to agree to each other within $\pm 30\%$.
- The model with different inventories can reproduce the BB enhancements in the profile but systematically underpredict VOCs observation by a factor of 2-6.

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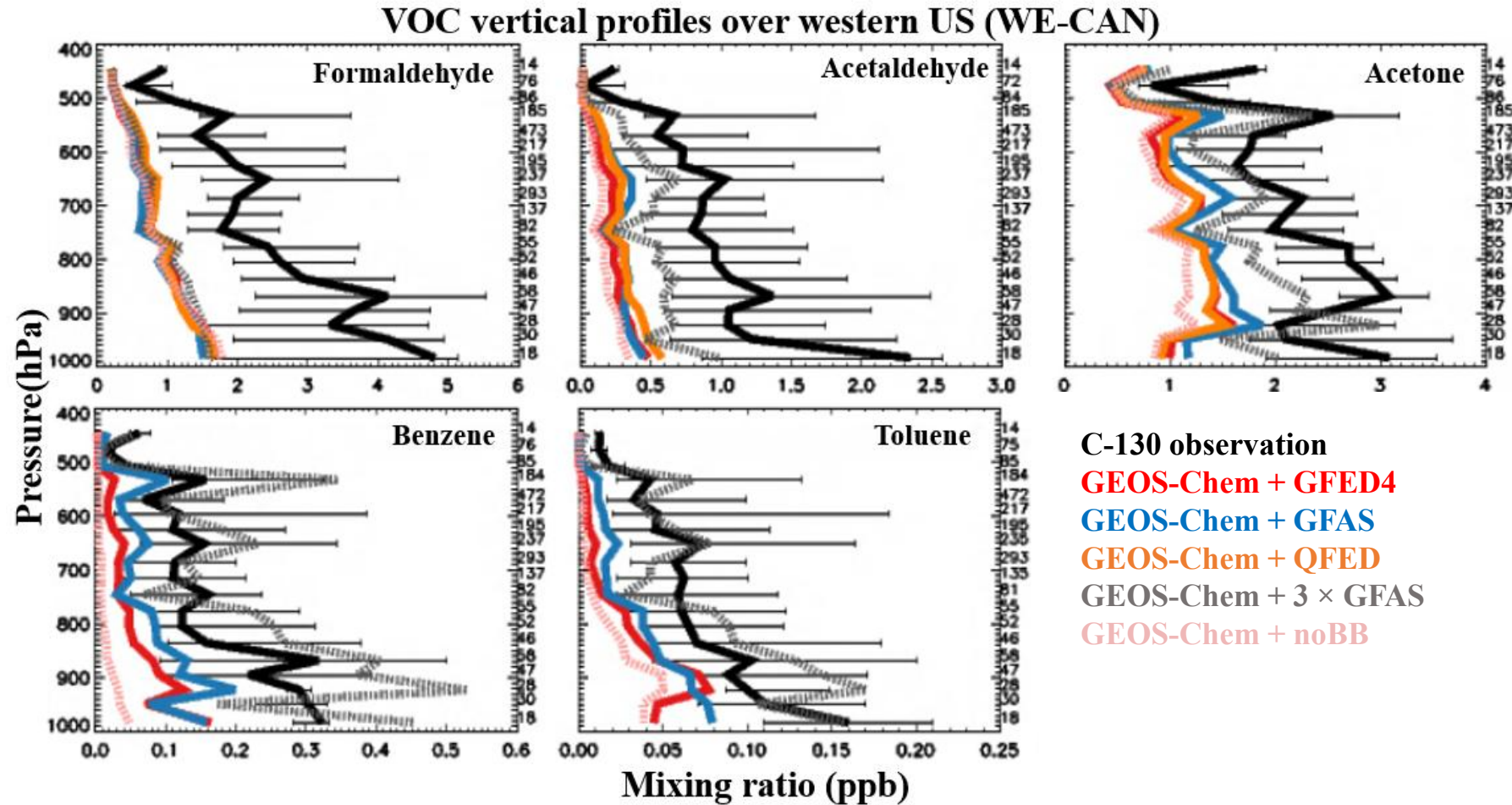


Is the negative bias from BB emissions? If so, how much BB emissions are underestimated? And why?



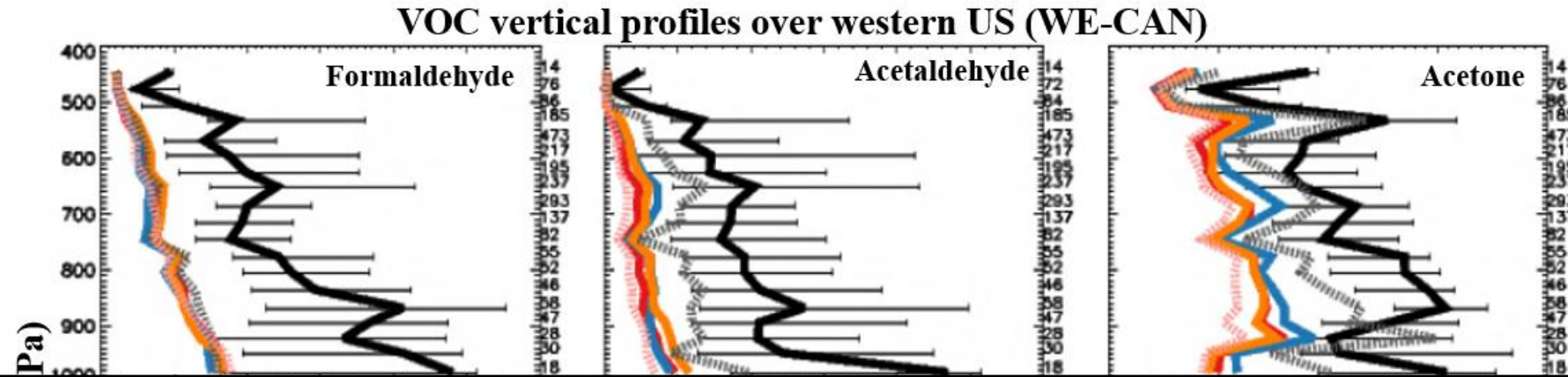
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Model underestimates BB emission and miss secondary production source for OVOCs

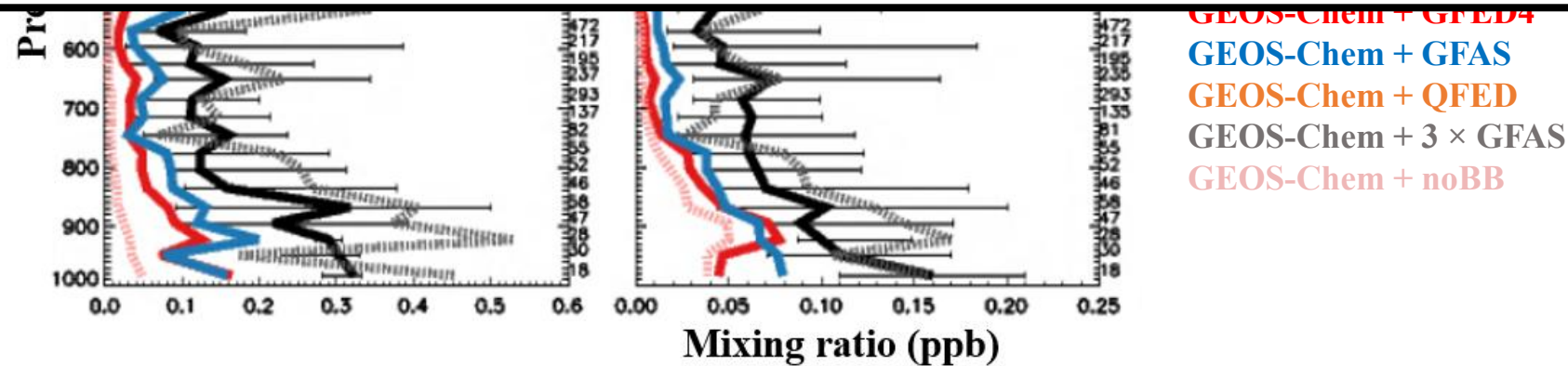


- Model underestimates the BB emission by a factor of 3-5.
- For primary VOCs, tripling BB emissions agree to the observation within measurement uncertainty.
- For OVOCs, there are likely missing photochemical production sources in the model.

Model underestimates BB emission and miss secondary production source for OVOCs



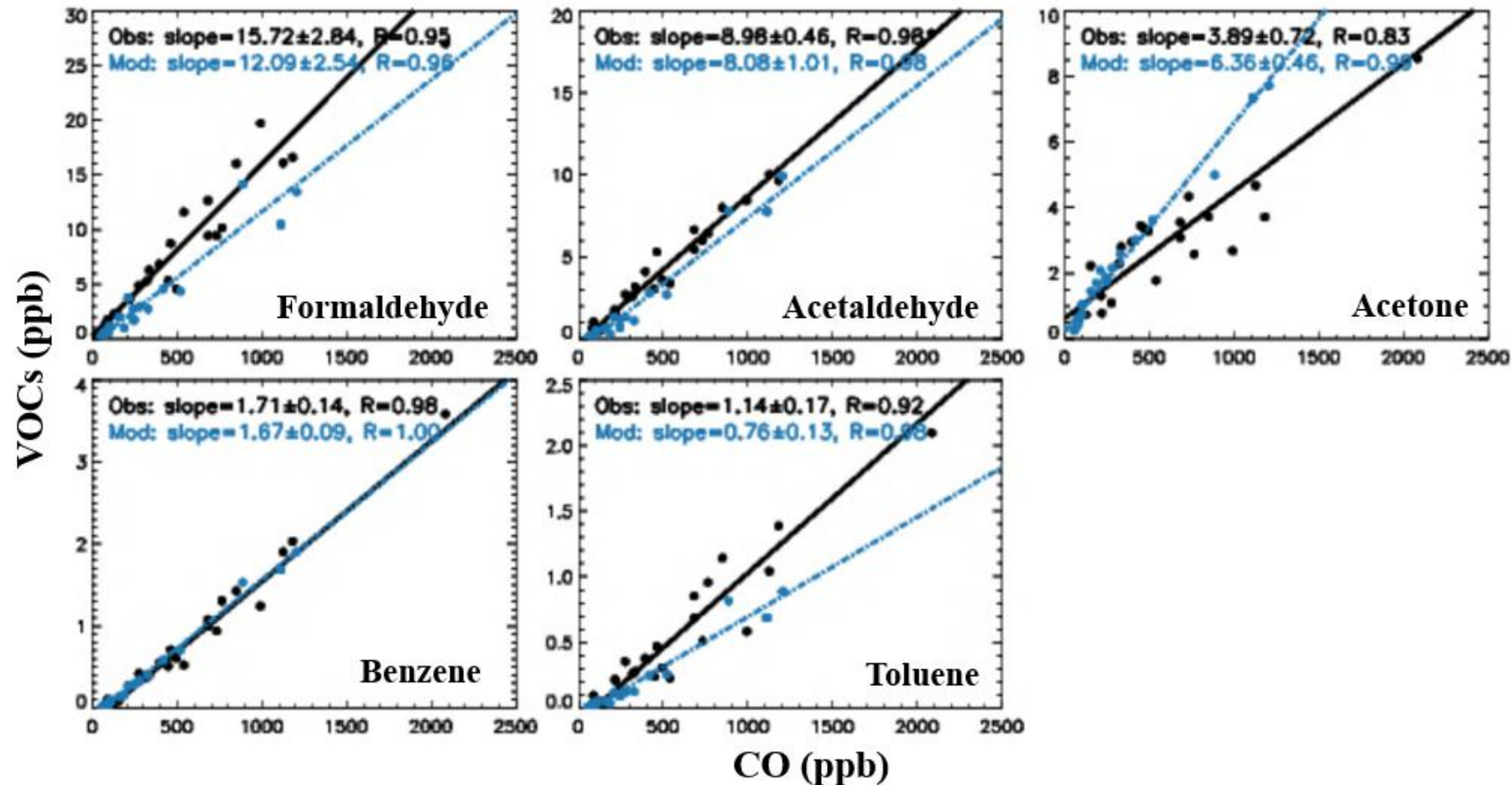
why does model underpredict BB emission?



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VOC emission ratio and fire detection are significant model errors in the GEOS-Chem

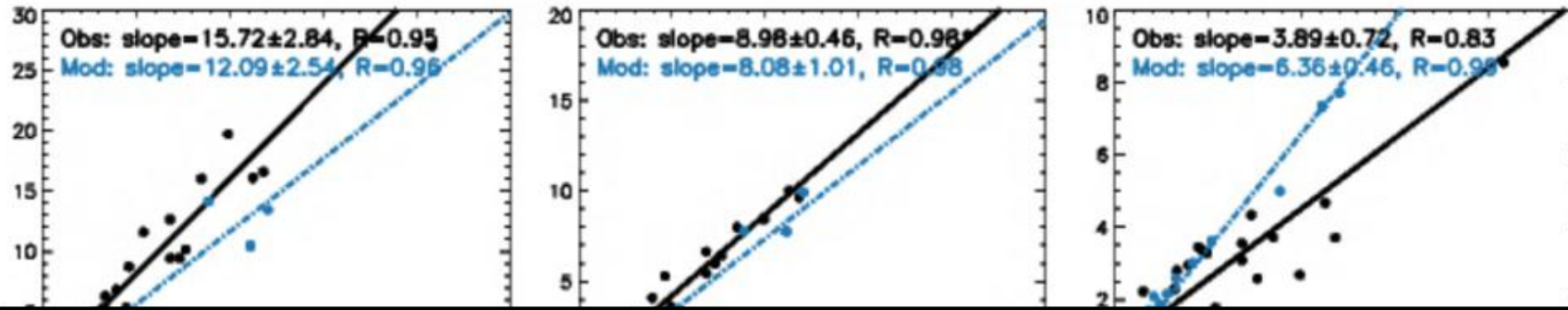
VOC vs CO emission ratio (WE-CAN)



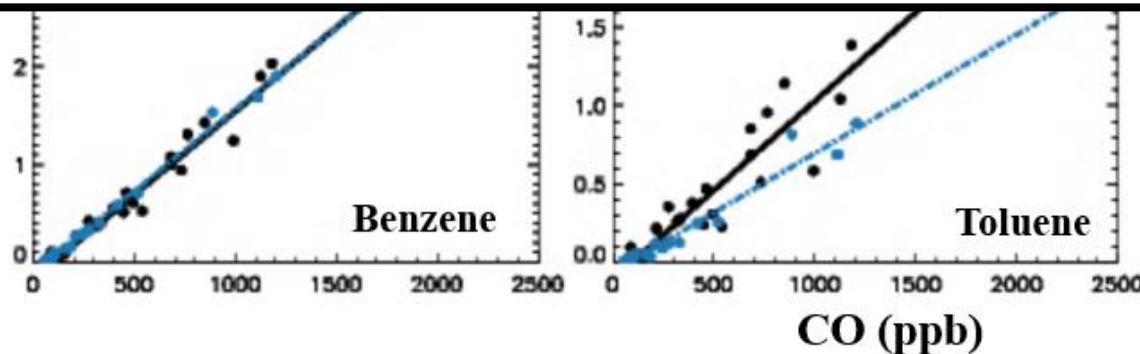
- Emission ratio test: **the best representation (GFAS) can agree observed ER within $\pm 30\%$ except acetone** (similar conclusion is found in FIREX-AQ).
- Fire detection experiment suggests the all 3 inventories can detect ~ 30 fires we sampled during WE-CAN.

VOC emission ratio and fire detection are significant model errors in the GEOS-Chem

VOC vs CO emission ratio (WE-CAN)



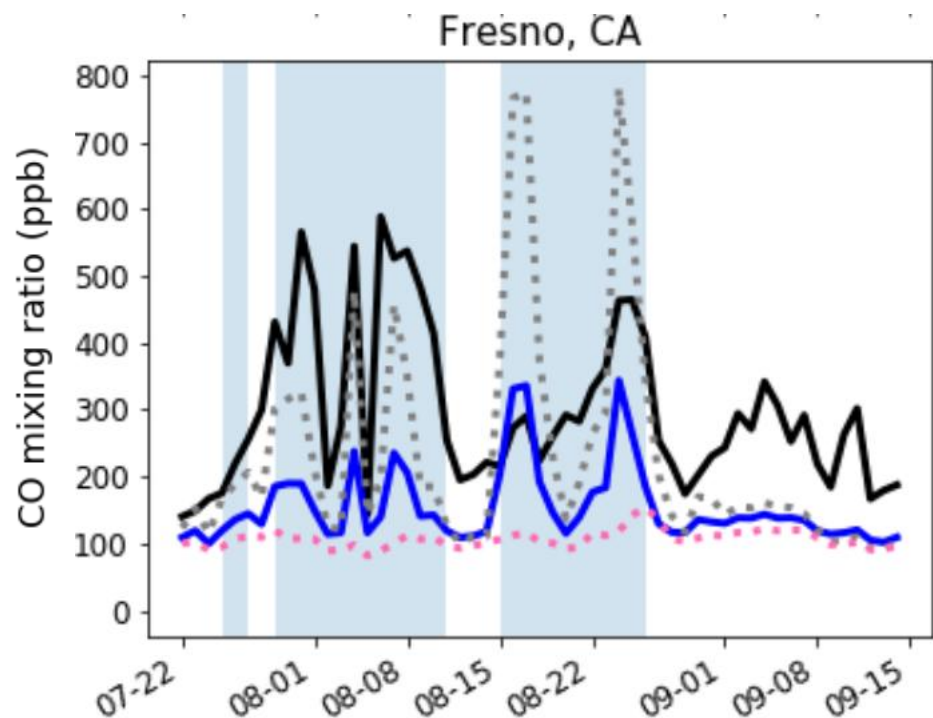
The BB underestimation is driven the dry matter underestimation.



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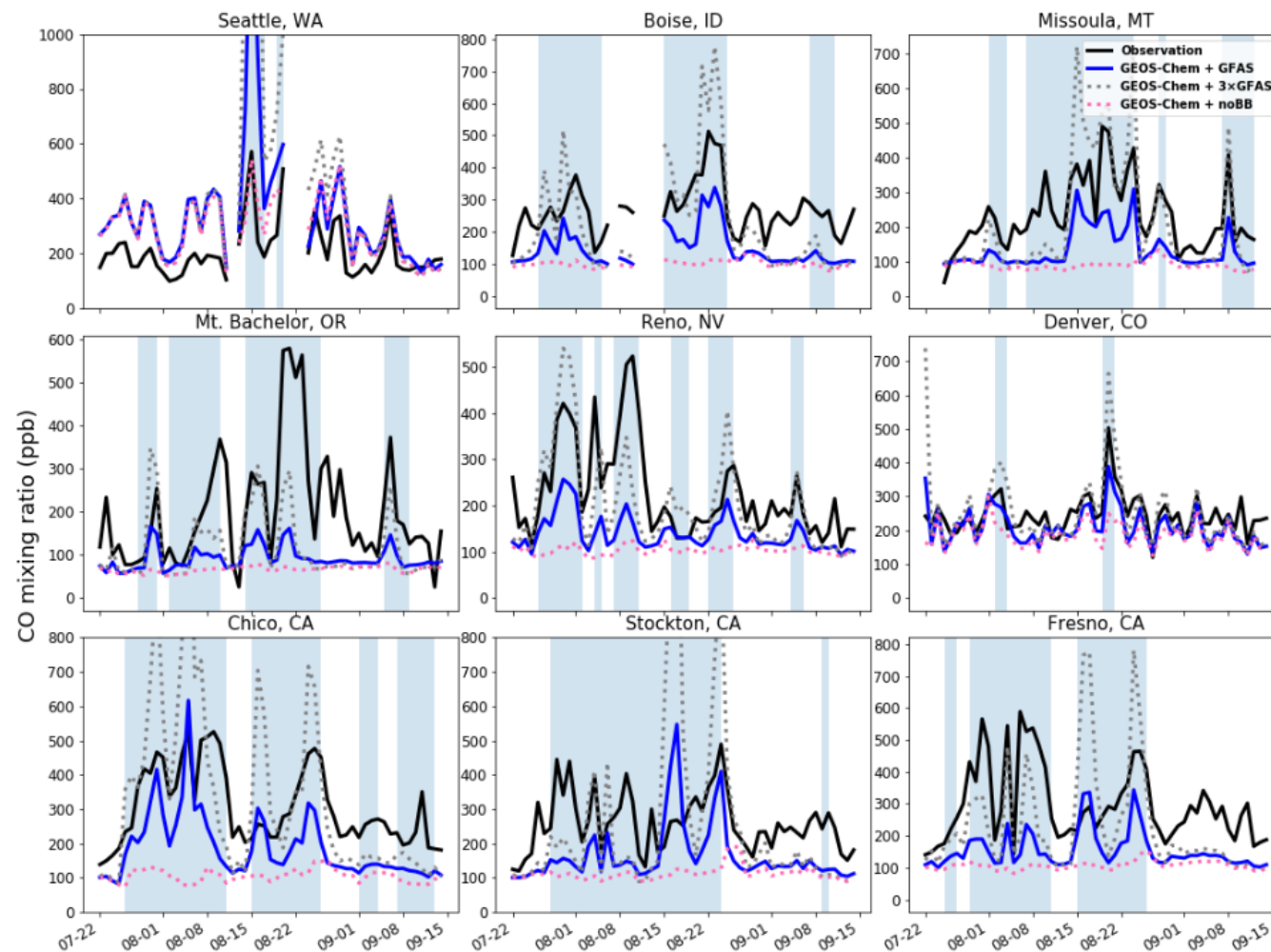
- The **base model** (GEOS-Chem + GFAS) does **underestimate** BB emissions across western US.
- The **$3 \times$ GFAS** simulation **systematically improves the model** mean bias across western US for 7 fire-influenced sites without degrading correlation coefficients with observations.

CO mixing ratios time series (WE-CAN)



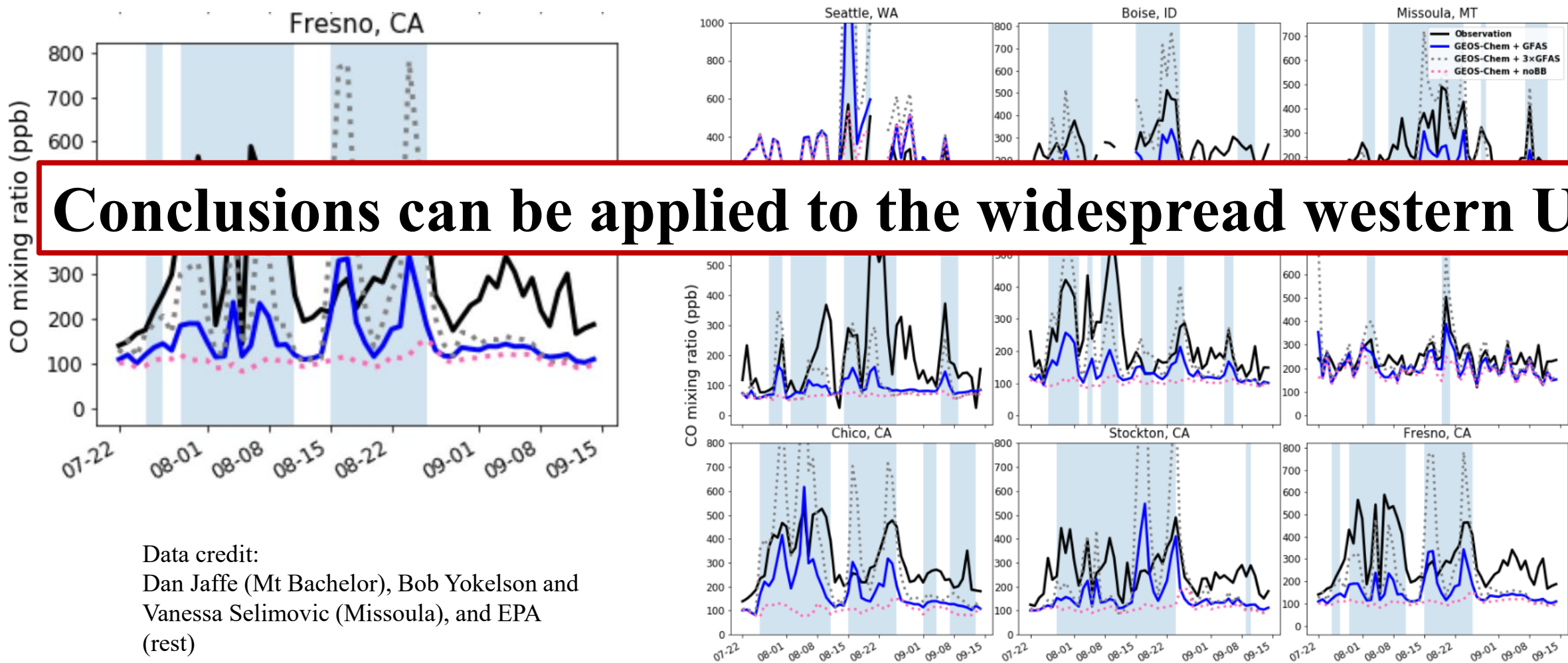
Data credit:

Dan Jaffe (Mt Bachelor), Bob Yokelson and Vanessa Selimovic (Missoula), and EPA (rest)

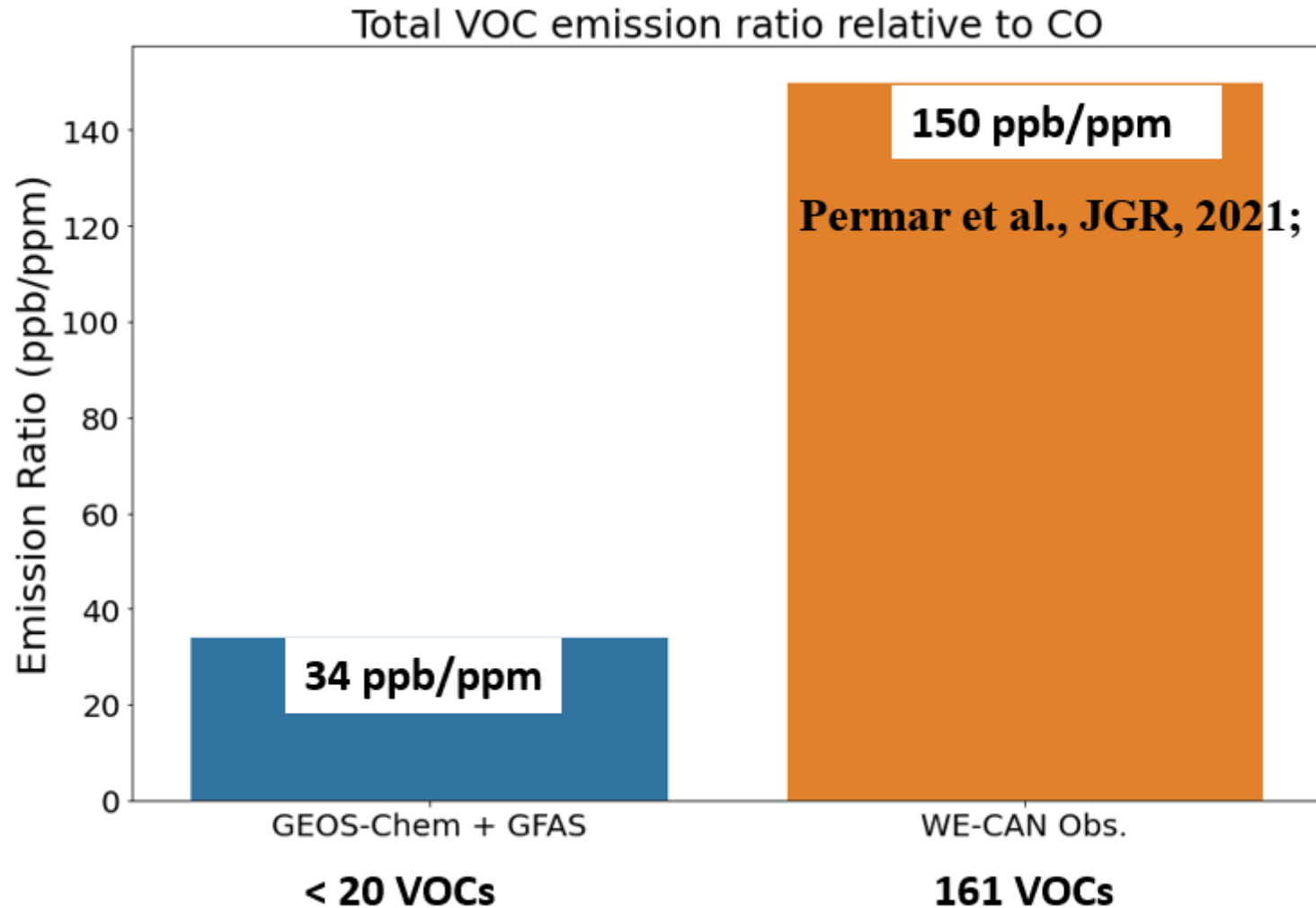


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CO mixing ratios time series (WE-CAN)



BB emission is far more important than the current understanding



- **Implication - VOCs emission from wildfire can play a more important role among the primary emissions (30-60% of the total; 1-8% of total before), considering the model error and unimplemented VOCs (see figure).**

Summaries

Conclusions:

- Three commonly-used BB inventories underestimate the BB emissions due to underprediction of dry matter burnt.
- Tripling the BB emission ($3 \times$ GFAS) systematically agree to observations for primary VOCs. However, model misses photochemical production source for OVOCs.
- BB emissions can play a far more important role than previous understanding.

Future work and preliminary results:

- Figure out the **missing chemistry for formaldehyde** (ongoing, C₂H₄ related).
- Improving the VOC representation regarding incomplete implemented VOC and unclear chemistry for **Furans and other VOCs** (Coggon et al., 2019; Permar et al., in prep).



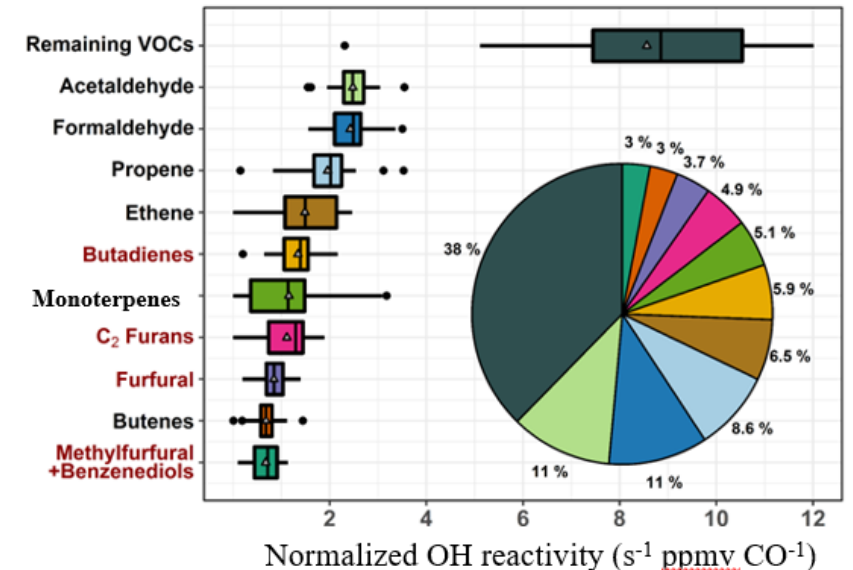
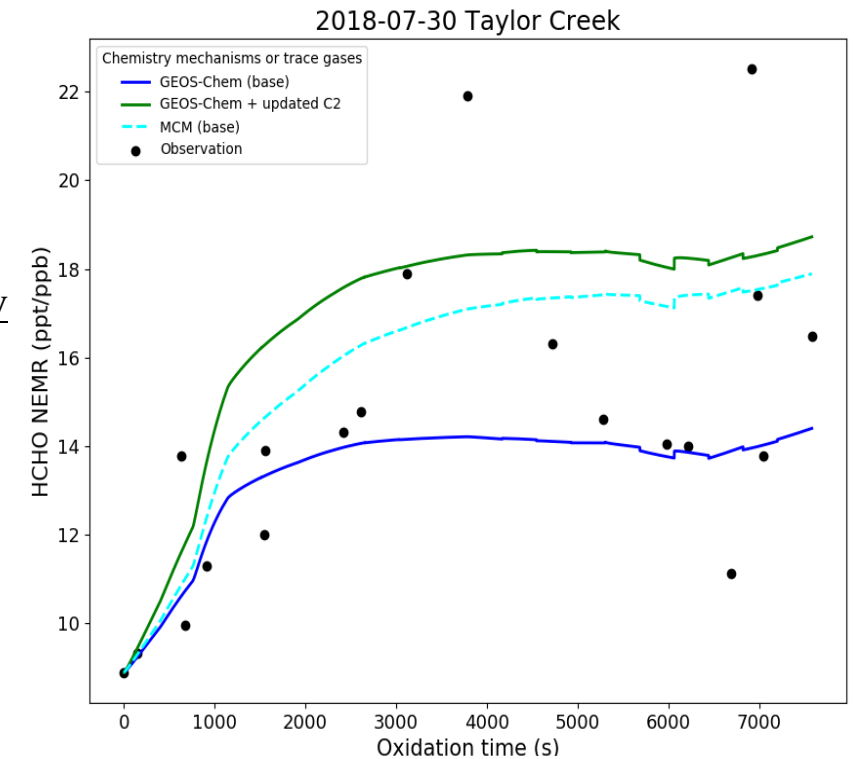
The draft is currently ready to share!!!



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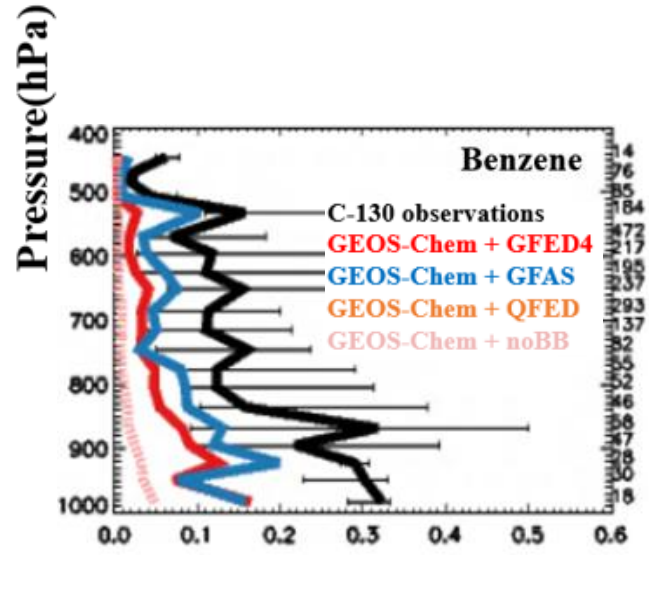
Twitter: @Lixu_Jin



Permar et al. (in review)

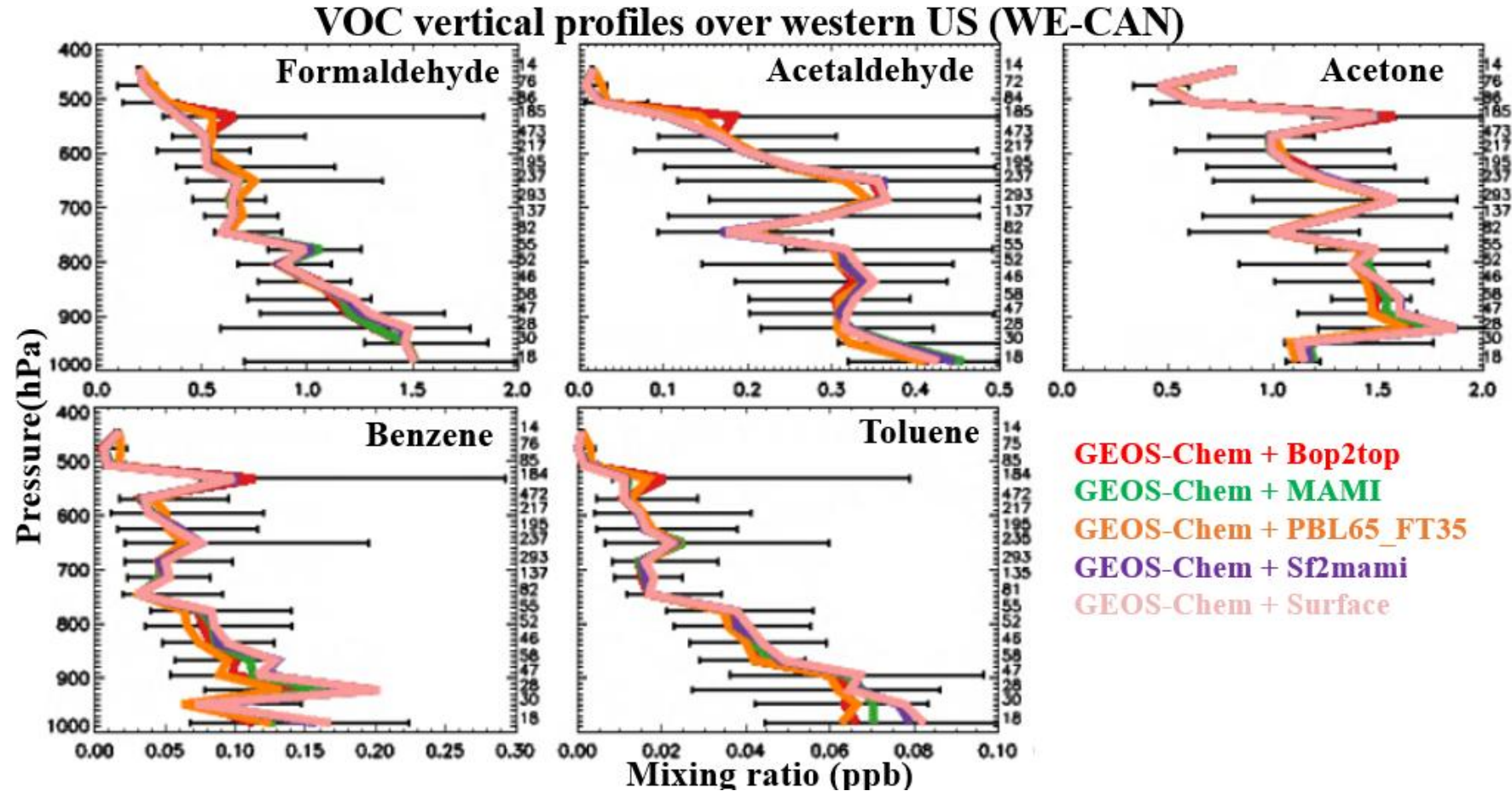
The model can predict the BB enhancements, but the emission size may be underestimated

VOC vertical profiles over western US (WE-CAN)



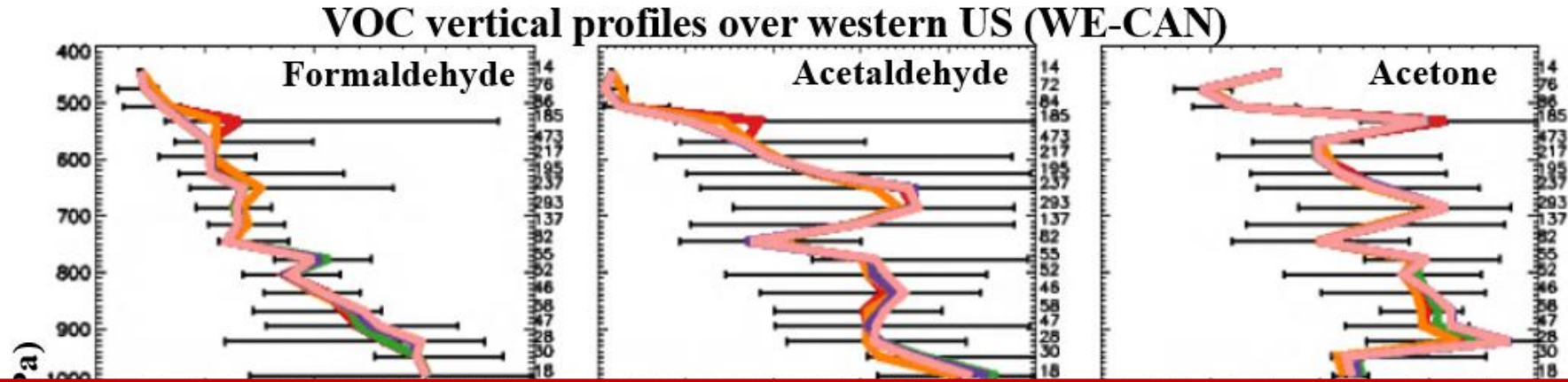
- The model with different inventories suggest similar enhancements and tend to agree to each other within $\pm 30\%$.
- Inventory experiments can reproduce the BB enhancements in the profile but systematically underpredict VOCs observation by a factor of 2-6.

GEOS-Chem is not sensitive to BB injections from the WE-CAN observation constraints

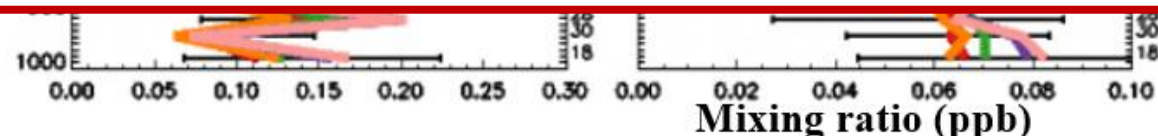


- Injection experiments tend to agree to each within 10%. **GEOS-Chem is not sensitive to injection in the western US wildfires.**
- However, the choice of plume injection heights can still be **important for secondary productions in downwind areas**. For example, daily mean ozone concentrations vary by up to 14 % or 4 ppb at surface in our sensitivity tests using GFAS (not shown).

GEOS-Chem is not sensitive to BB injections from the WE-CAN observation constraints

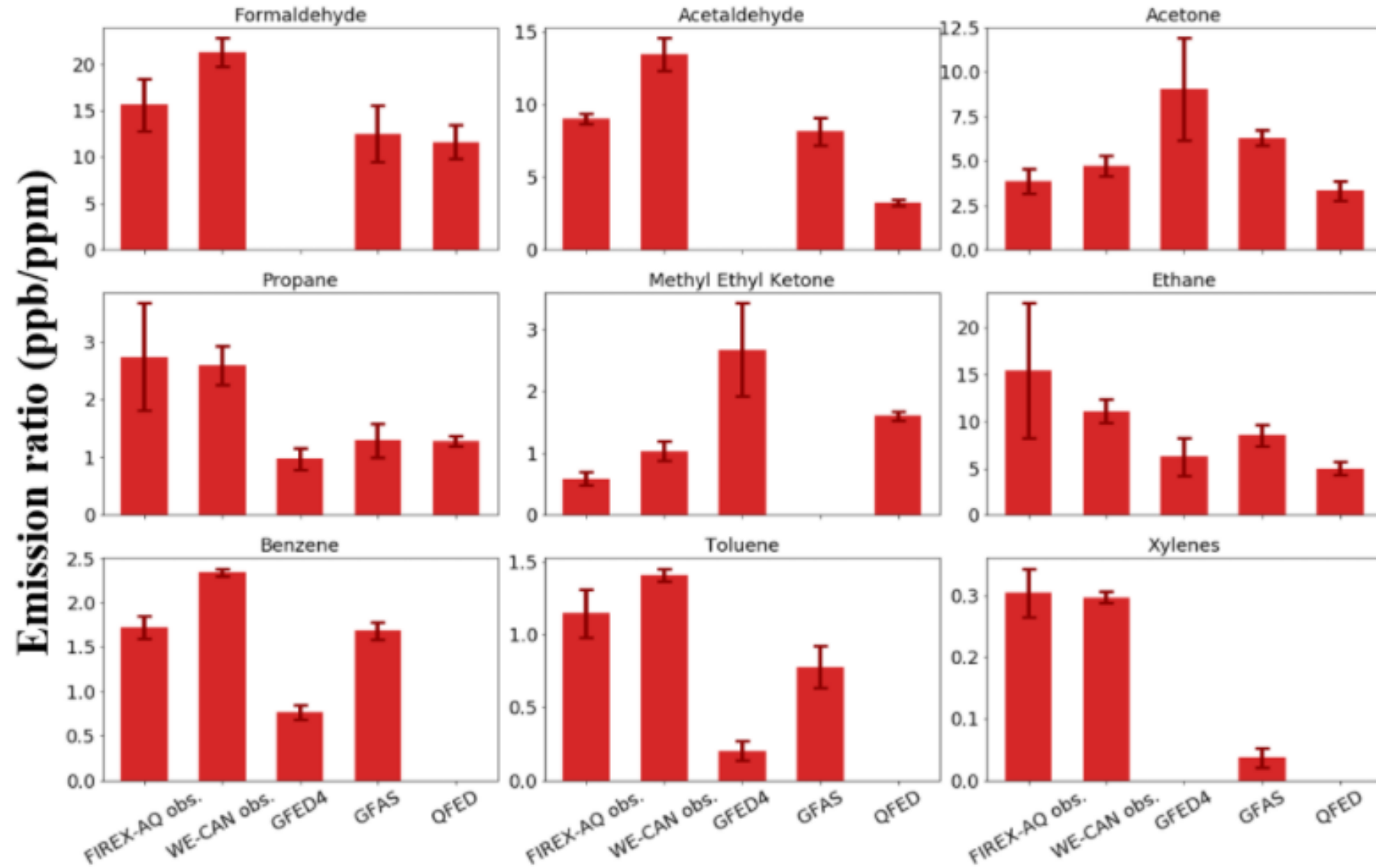


Injection is not a significant model error for WE-CAN. So why does model underpredict BB emission?

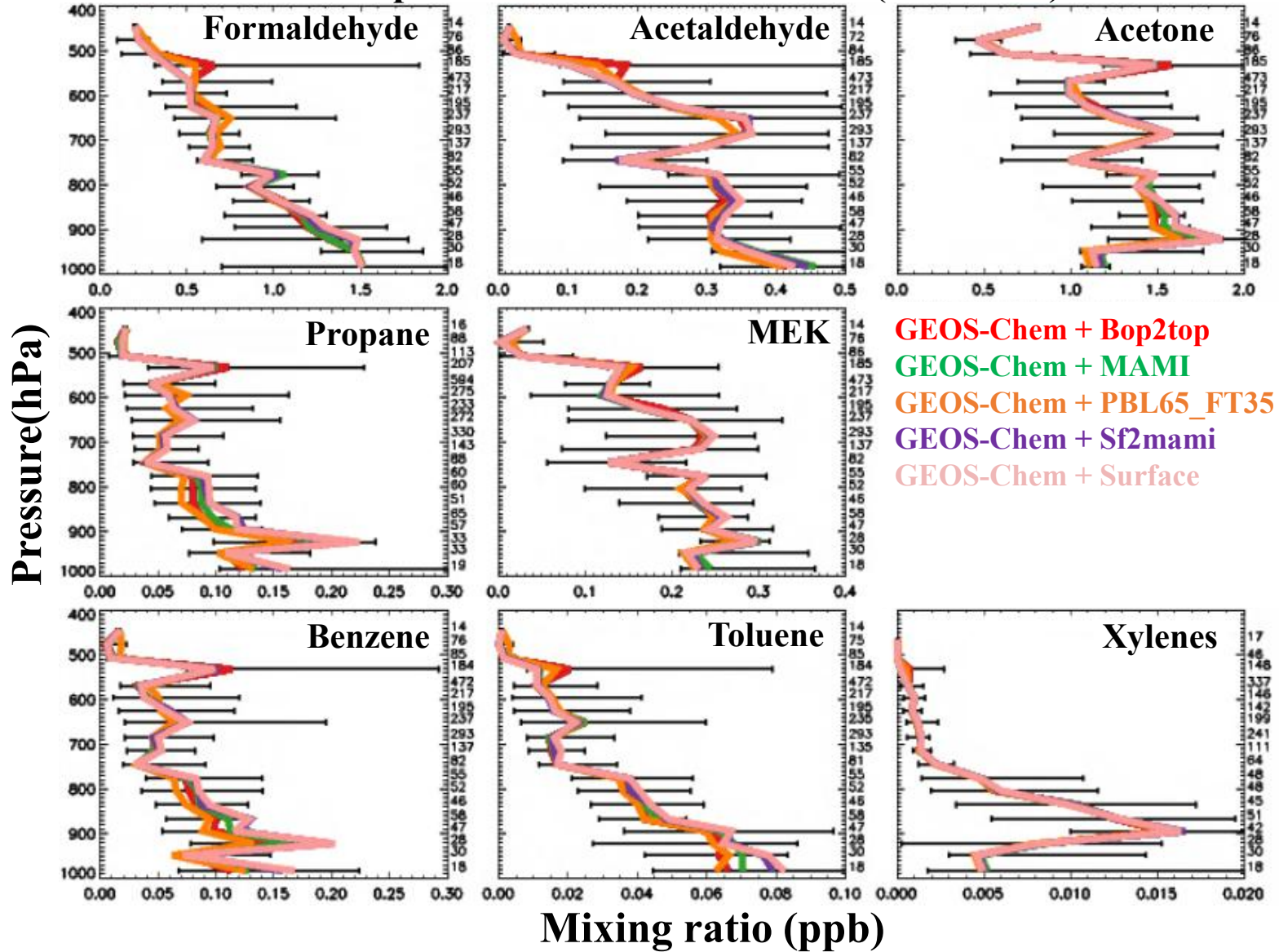


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VOC vs CO emission ratio



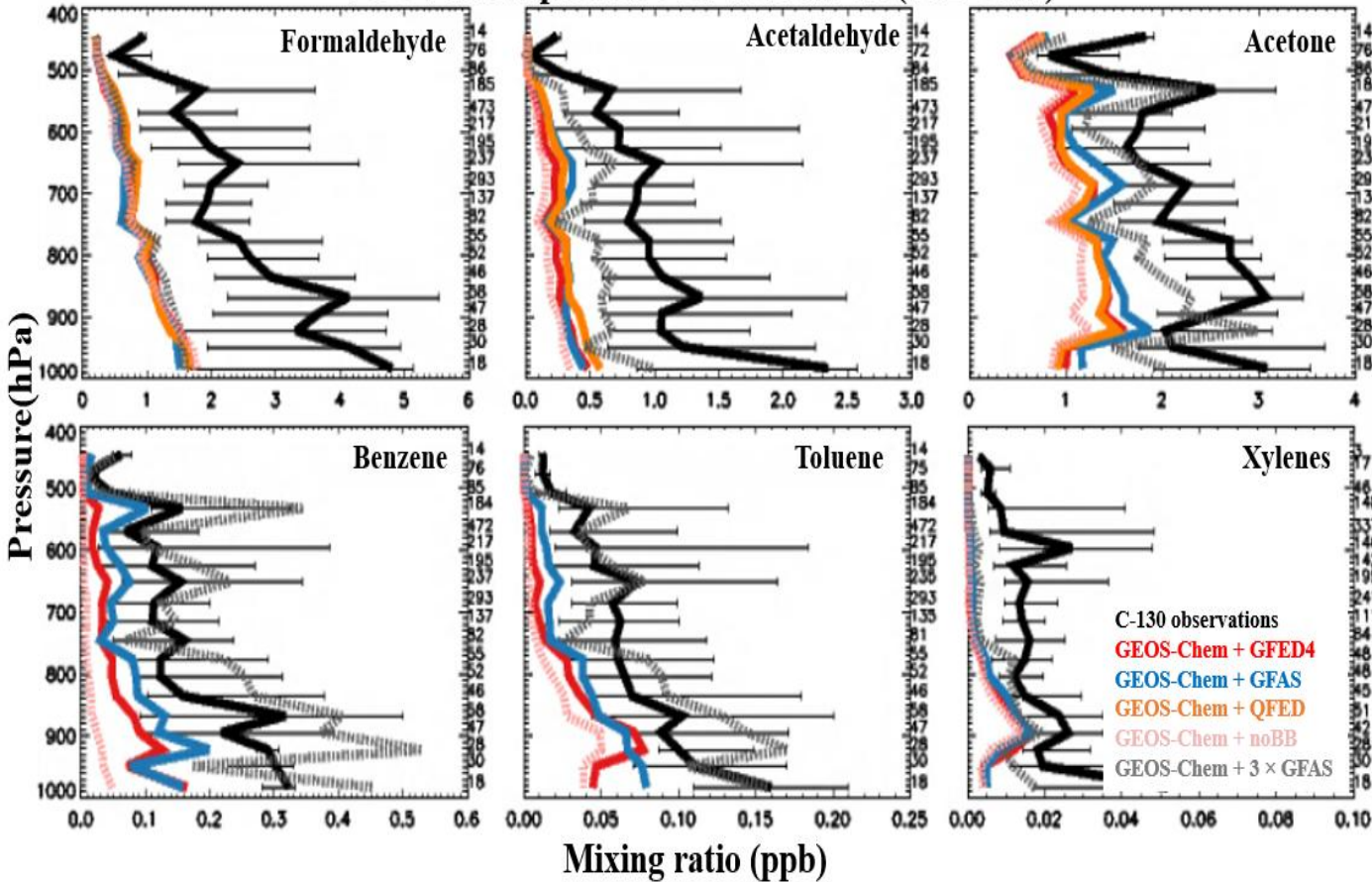
VOCs vertical profiles over the western US (WE-CAN)



- GEOS-Chem is not sensitive to injection in the western US wildfires.

The model underpredicts the BB emissions by a factor of 3-5 (Q1 and Q2)

VOC vertical profiles over western US (WE-CAN)



- Simple math is conducted to derive the underpredicted BB:
- BB impacts = full datasets – least fire conditions
- Underestimated BB impacts = BB impacts from observation – from model

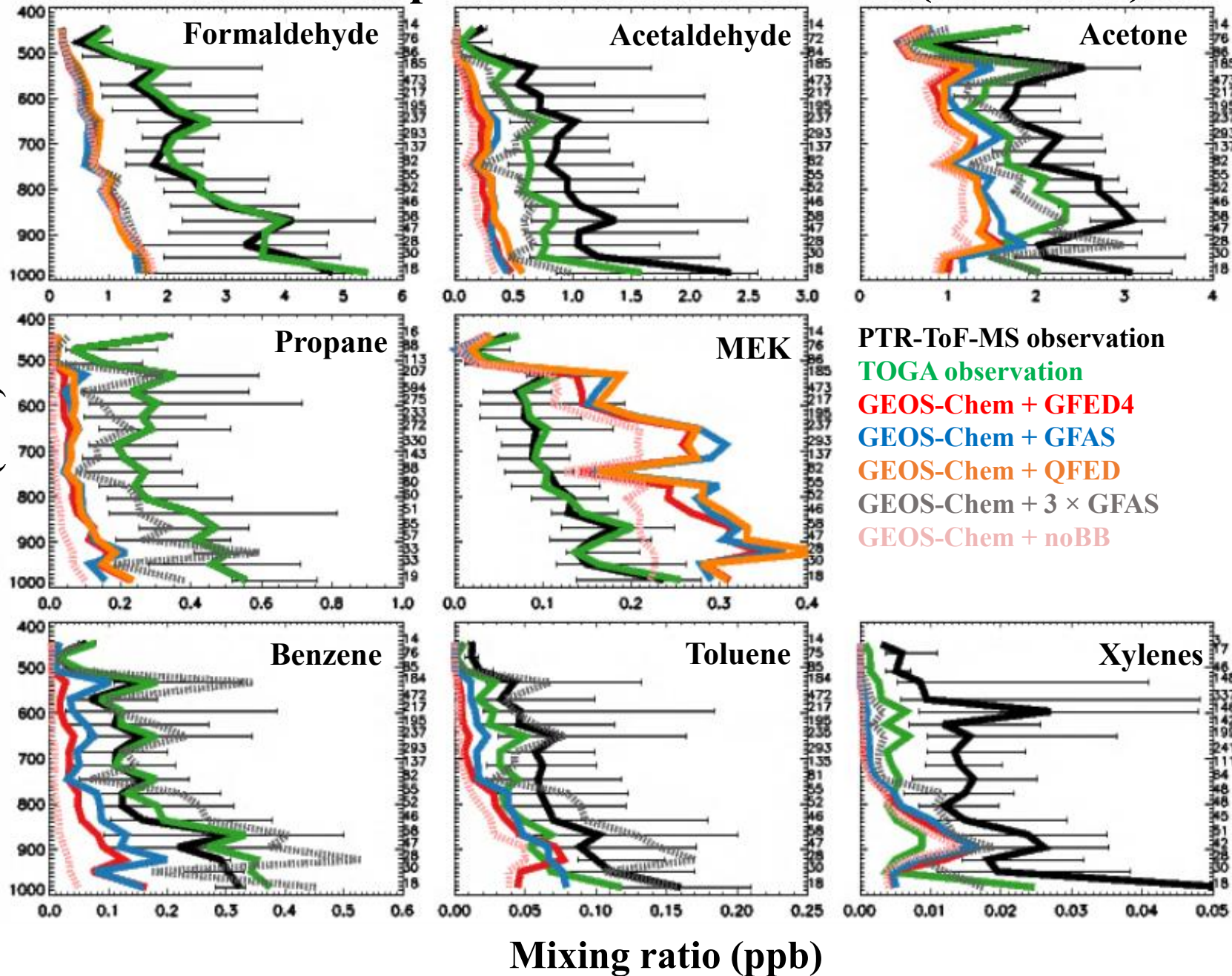
⇒ Model underestimates by a factor of 3-5 for primary trace gases (similar conclusion is gained for the 2019 FIRE-AQ)

⇒ Tripling the BB emissions in the model agree observed VOCs within observation uncertainty except xylenes and OVOCs (formaldehyde here).

⇒ **Why are BB emissions underestimated?**
(e.g., emission factors, fuel burned, fire detections, and injections?)

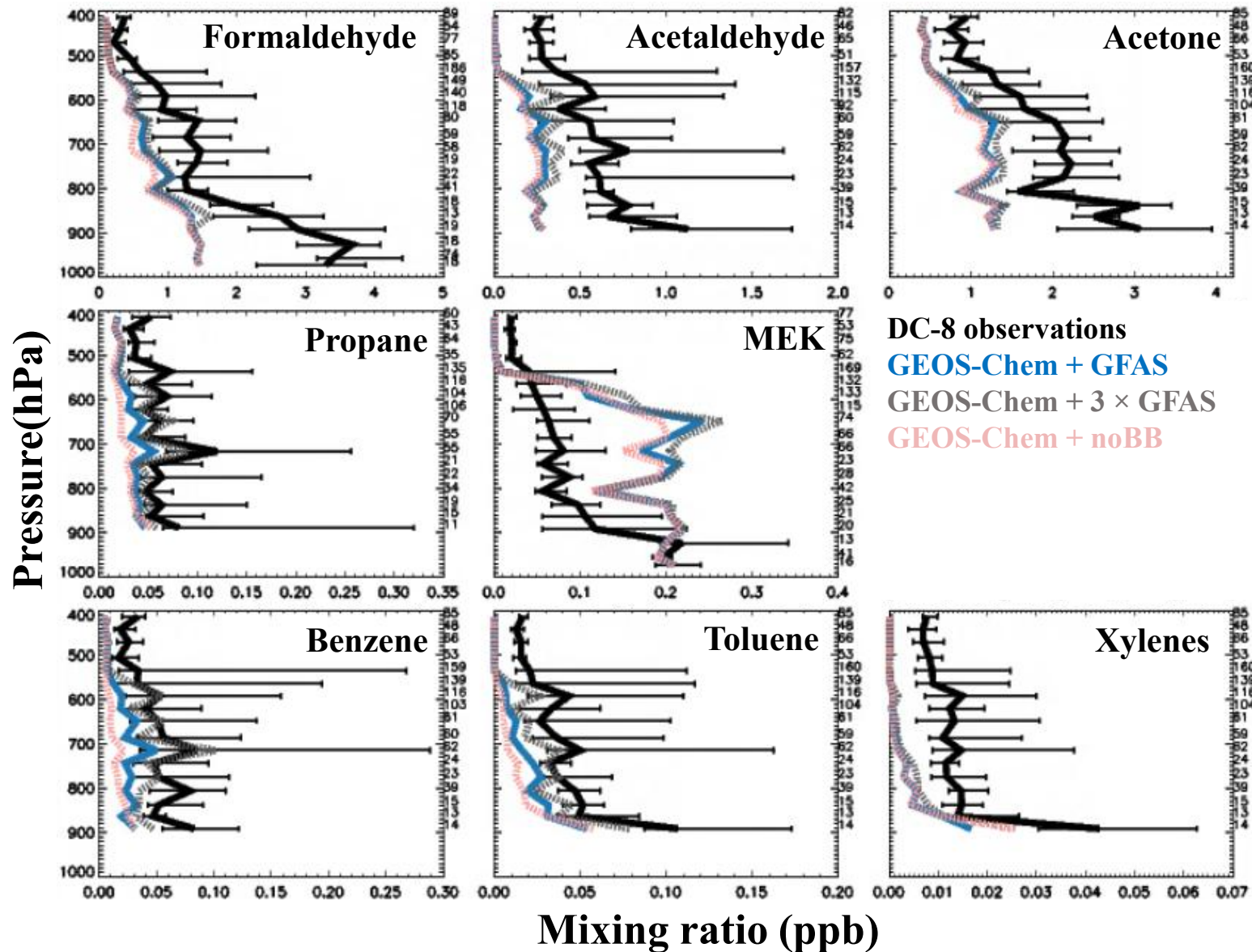
VOC vertical profiles over western US (WE-CAN)

Pressure(hPa)



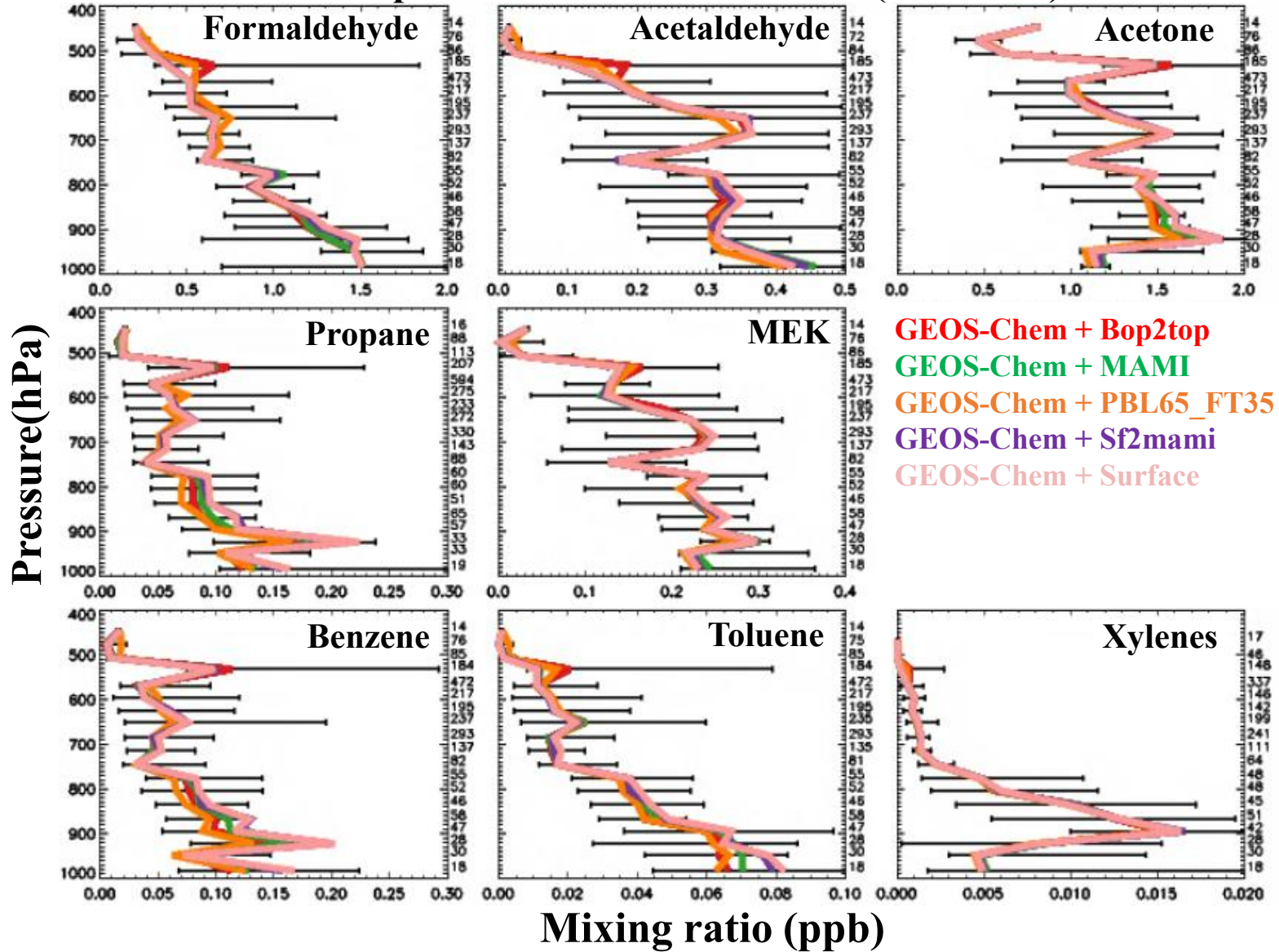
- Tripling BB emissions agree the observations with measurement uncertainty except formaldehydes, xylenes, MEK and lumped $C \geq 3$ aldehydes (open to discuss if anyone is interested in)

VOCs vertical profiles over the western US (FIREX-AQ)



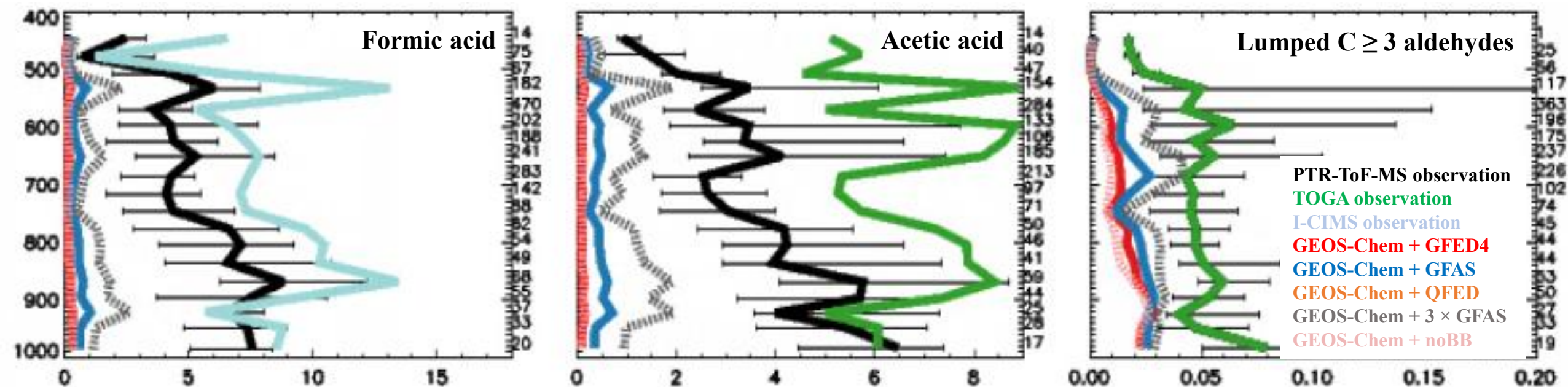
- **Similar conclusion supported by FIREX-AQ:** GEOS-Chem + GFAS can reproduce the BB enhancements in the profile but systematically underpredict BB-related source by a factor of 3-6 for primary VOCs (except MEK; a factor of 2-5 in the WE-CAN).

VOCs vertical profiles over the western US (WE-CAN)



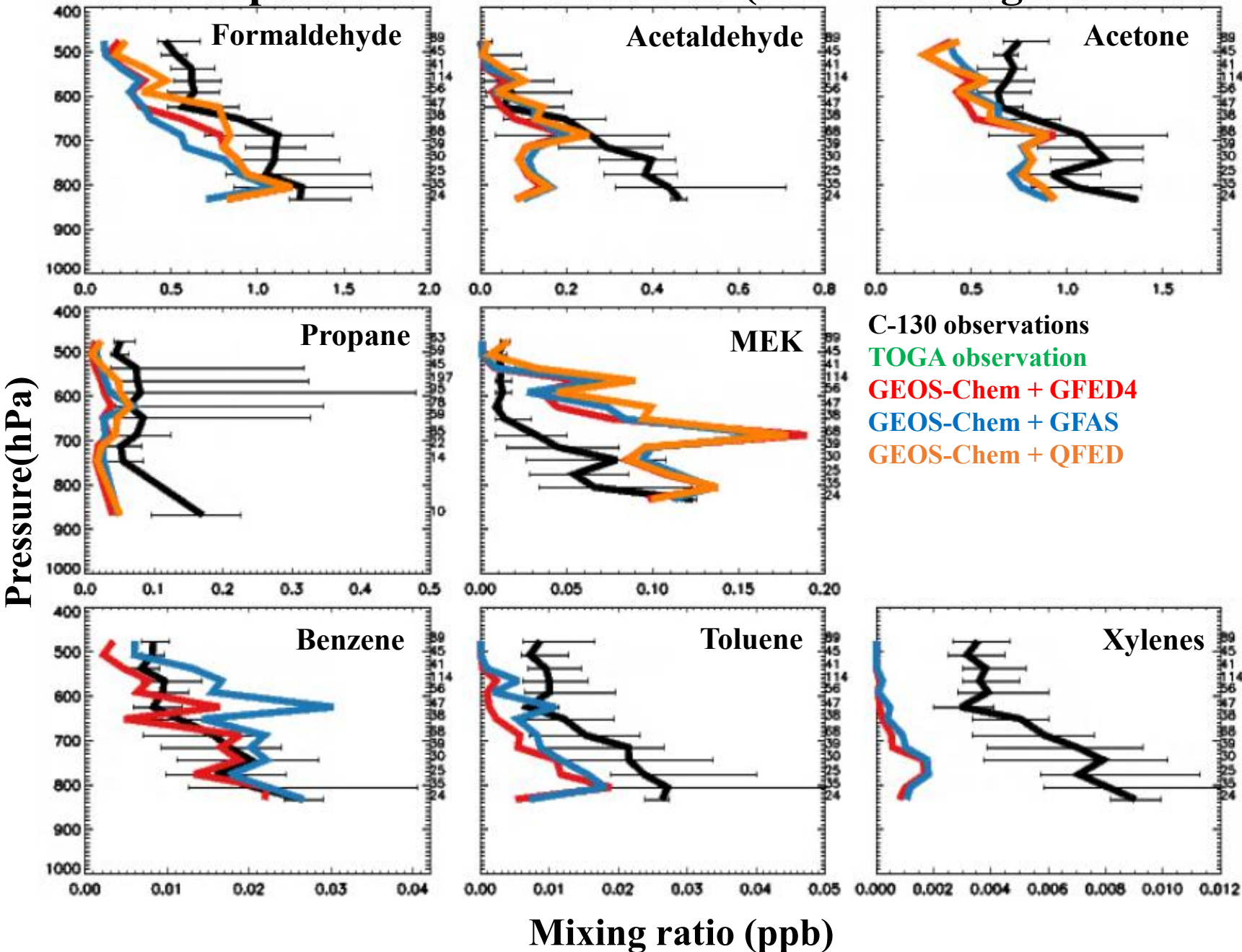
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VOC vertical profiles over western US (WE-CAN)

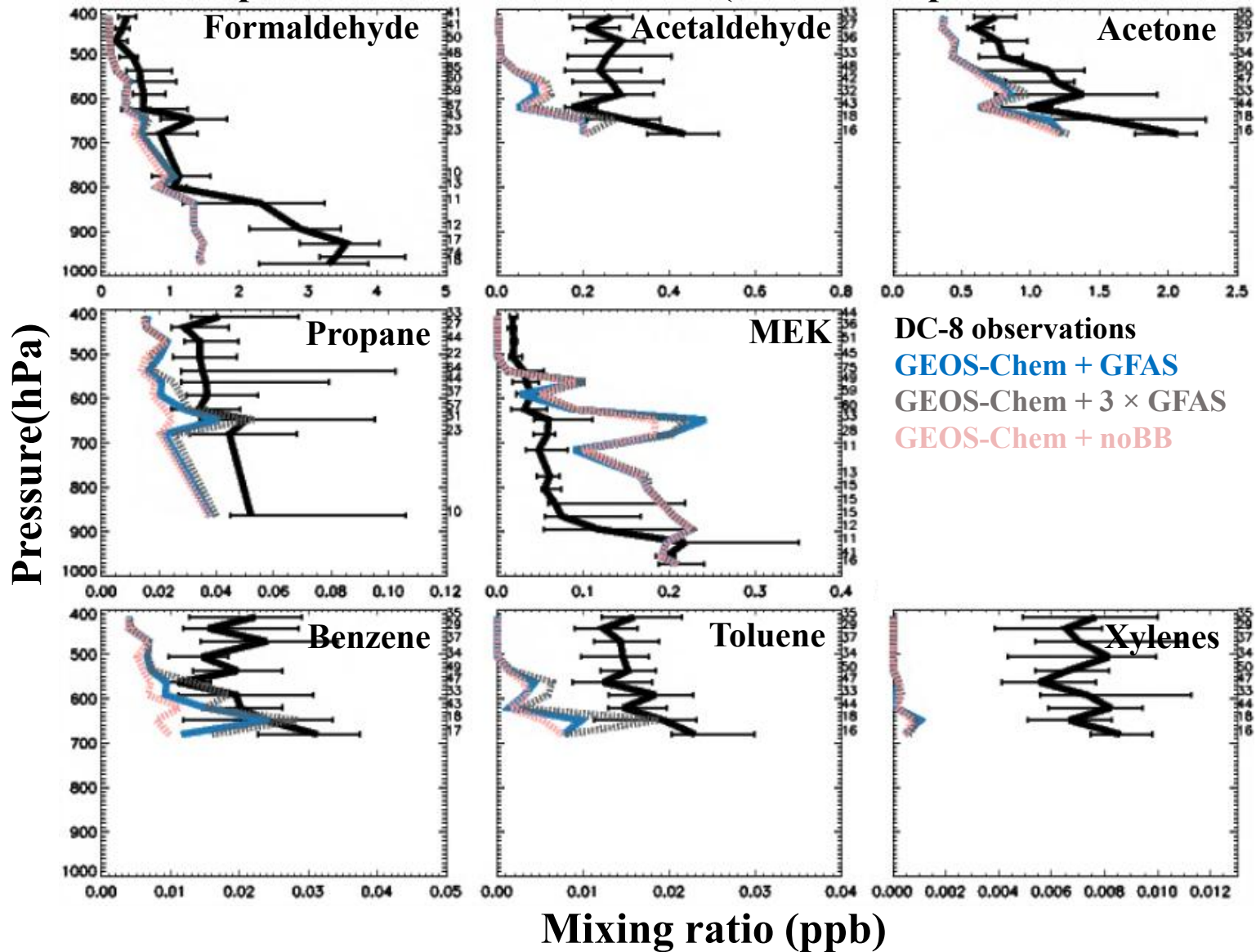


- Missing photochemical production source for formic acid, acetic acid, and lumped C ≥ 3 aldehyde

VOC vertical profiles over western US (least-fires region for WE-CAN)

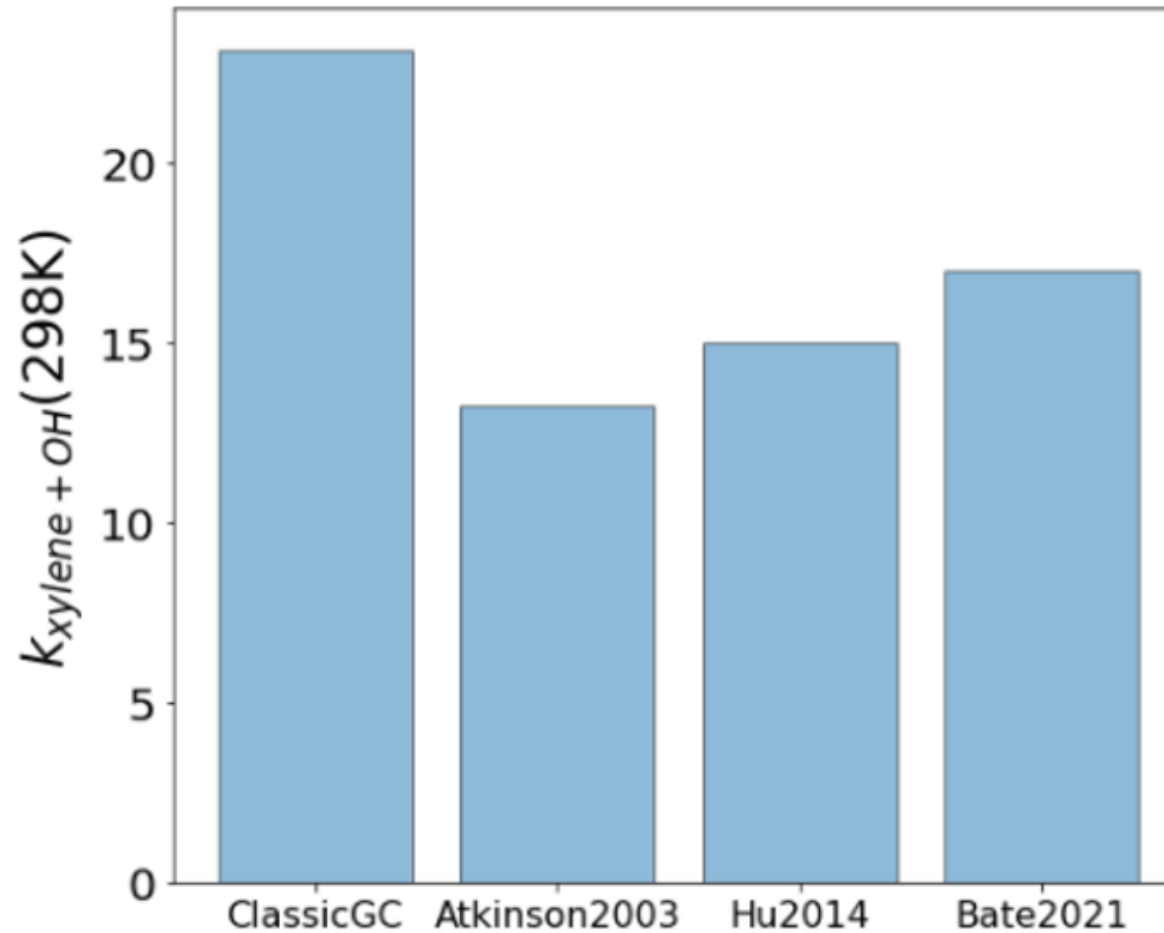


VOCs vertical profiles over the western US (least BB-impacted for FIREX-AQ)



There are problems beyond BB emissions as suggested by the relatively clean environment: missing HCHO, acetaldehyde, propane, benzene, and toluene.

Rate constant of xylenes + OH



Rate constants for the gas-phase reactions of the OH radical with xylenes -> **Koh in GCv12 is too high, introducing a shorter lifetime and thus lower mixing ratio.**